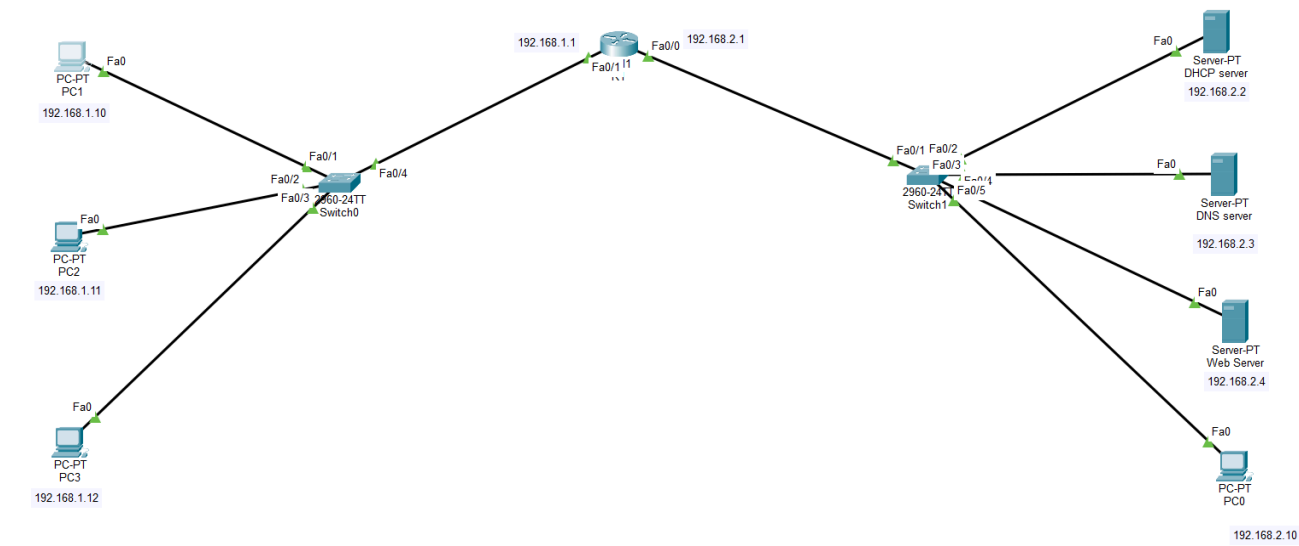


Web-Server Construction Project Architecture:



DHCP Config:

DHCP server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.2.1

DNS Server: 192.168.2.3

Start IP Address: 192.168.2.10

Subnet Mask: 255.255.255.0

Maximum Number of Users: 50

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.2.1	192.168.2.3	192.168.2.10	255.255.255.0	50	0.0.0.0	0.0.0.0
serverPool1	192.168.1.1	192.168.2.3	192.168.1.10	255.255.255.0	50	0.0.0.0	0.0.0.0

DNS Config:

DNS server

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DNS

DNS Service

On

Off

Resource Records

Name

prasanna.com

Type

A Record

Address

192.168.2.4

Add

Save

Remove

No.	Name	Type	Detail
0	prasanna.com	A Record	192.168.2.4

DNS Cache

Top

Web-Server Config:

Web Server

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration X

IP Configuration

DHCP

Static

IPv4 Address

192.168.2.4

Subnet Mask

255.255.255.0

Default Gateway

192.168.2.1

DNS Server

192.168.2.3

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address

FE80::260:47FF:FE8C:159A

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

HTTP

On

Off

HTTPS

On

Off

File Manager

	File Name	Edit	Delete
1	architecture.jpg		(delete)
2	copyrights.html	(edit)	(delete)
3	cscoptlogo177x111.jpg		(delete)
4	dhcp.jpg		(delete)
5	dns.jpg		(delete)
6	helloworld.html	(edit)	(delete)
7	image.html	(edit)	(delete)
8	index.html	(edit)	(delete)

New FileImport

Output: (PC1- Desktop - Web Browser-prasanna.com)

PC1

PhysicalConfigDesktopProgrammingAttributes

Web Browser

<>URLhttp://prasanna.comGoStop

 **CCNA Webserver Construction Project**

Final Execution & Documentation

Student: PRASANNA VELAN V

Trainer: Aadithya

Course: CCNA

Institute: Besant Technologies - Jayanagar

Project: CCNA Webserver Construction Project

Status: Execution Successful ✓

 **Project Overview**

This project represents a complete functioning network built inside Cisco Packet Tracer with support for:

- Web Server Hosting
- DNS Domain Resolution
- DHCP Address Allocation
- Routing Between Subnets
- Client Network Communication

☐ Top

PC1

PhysicalConfigDesktopProgrammingAttributes

Web Browser

<>URLhttp://prasanna.comGoStop

Network Architecture

The diagram below shows the implemented topology:

```
graph LR
    R1[Router R1] --- S1[Switch S1]
    R1 --- S2[Switch S2]
    S1 --- PC1[PC1]
    S1 --- PC2[PC2]
    S1 --- PC3[PC3]
    S1 --- DHCP[DHCP server]
    S2 --- DNS[DNS server]
    S2 --- WS[Web Server]
    S2 --- PC0[PC0]
```

Subnet Allocation & Device Roles

Device	Role	IP / Service
Router	Gateway	192.168.1.1 / 192.168.2.1
DNS Server	FQDN Resolver	192.168.2.3
DHCP Server	IP Allocation	192.168.2.2
Web Server	HTTP Hosting	192.168.2.4

Top

PC1

PhysicalConfigDesktopProgrammingAttributes

Web Browser

<>URLhttp://prasanna.comGoStop

Subnet Allocation & Device Roles

Device	Role	IP / Service
Router	Gateway	192.168.1.1 / 192.168.2.1
DNS Server	FQDN Resolver	192.168.2.3
DHCP Server	IP Allocation	192.168.2.2
Web Server	HTTP Hosting	192.168.2.4
PC1	Client	192.168.1.10 (DHCP)
PC2	Client	192.168.1.11 (DHCP)
PC3	Client	192.168.1.12 (DHCP)
PC0	Client	192.168.2.10 (DHCP)

DHCP Server Implementation

DHCP server issues IPs automatically to all client devices inside the network. This eliminates manual static IP assignment.

- DHCP Pool Created
- DNS Assigned Automatically
- Default Gateway Assigned

DHCP server

PhysicalConfigServicesDesktopProgrammingAttributes

Top

PC1

Physical Config Desktop Programming Attributes

Web Browser

URL http://prasanna.com

Go Stop

DHCP Server Implementation

DHCP server issues IPs automatically to all client devices inside the network. This eliminates manual static IP assignment.

- DHCP Pool Created
- DNS Assigned Automatically
- Default Gateway Assigned

DHCP server

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface FastEthernet0 Service On Off

Pool Name serverPool

Default Gateway 192.168.2.1

DNS Server 192.168.2.3

Start IP Address : 192 168 2 10

Subnet Mask: 255 255 255 0

Maximum Number of Users : 50

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.2.1	192.168.2.3	192.168.2.10	255.255.255.0	50	0.0.0.0	0.0.0.0
serverPool1	192.168.1.1	192.168.2.3	192.168.1.10	255.255.255.0	50	0.0.0.0	0.0.0.0

Top

PC1

Physical Config Desktop Programming Attributes

Web Browser

URL http://prasanna.com

Go Stop

DNS Server Implementation

DNS server resolves the domain name of the hosted website to the Web Server's IP. Example Flow:

www.project.local → 192.168.2.4

DNS server

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DNS

DNS Service On Off

Resource Records

Name prasanna.com Type A Record

Address 192.168.2.4

Add Save Remove

No.	Name	Type	Detail
0	prasanna.com	A Record	192.168.2.4

